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Besvarelsen afleveres af

Phillip Hvidesten Trampedach
zct525@alumni.ku.dk

Eksamensadministratorer

Bedømmere

Mikkel Godt Gregersen

Eksaminator

mikkel.godt.gregersen@econ.ku.dk

Jens-Otto Skovgaard Jeppesen

Censor

jojtst@gmail.com

+4524945612

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Phillip H. Trampedach (ZCT525)

ESG-Scores and Debt-to-Equity Ratio

Does ESG-Score affect capital structure

Advisor: Mikkel Godt Gregersen

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Abstract

In this paper I seek to contribute to existing literature surrounding the topic of ESG-scores. In particular, I look into the relationship between ESG-scores and the debt-to-equity ratio for firms in the STOXX 600 index which is gathered from Refinitiv Eikon. The results are based on a sample of 483 firms. My main hypothesis state that high ESG-scores leads to higher debt-to-equity ratios. The analysis is based around an OLS regression with multiple control-variables and multiple GICS sectors as dummy variables to see if there is a relationship. I get that there is a positive relationship between the ESG-scores and the debt-to-equity ratio on a 10% significance level. This is based on the regression and on correlation coefficients. I therefore can't confirm the hypothesis on a 5% significance level, but I find evidence that supports it, as I find that there is a significance on a 10% level between the ESG-scores and the debt-to-equity ratio. My second hypothesis is about if the possible effect of ESG-scores on the debt-to-equity level varies by sector. I only use the correlation coefficients to test this. Optimally, I would have made a separate regression for each sector. I don't find any significant correlation between ESG-scores and debt-to-equity ratios in any sector, from which I currently dismiss the hypothesis. From this I conclude, that there is evidence which points in the direction of an overall positive relationship between ESG-scores and the debt-to-equity ratio. This means that there is evidence for which a higher ESG-score leads to more debt. I also conclude that there isn't a significant difference between which sector the firms participates in and the correlation between the ESG-score and the debt-to-equity ratio.

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1 Introduction

In recent times green investments have become more and more popular, as individuals and firms seek to help battle climate change. In this paper, I seek to investigate if ESG-scores has an effect on the debt-to-equity ratio for firms in the STOXX 600 index. I will do this by setting up an OLS regression with multiple-control variables, which will take into account already know relationships. This will separate the relationship between the ESG-score and the debt-to-equity ratio, from which this paper is based around. As a supplement to the OLS regression, I will also be calculating the Pearson correlation coefficient. This will be used to see if there is any significant correlations between the dependent variable and the ESG-score alongside the control-variables. The research is being limited to only consisting of European firms. This is done to give the paper a focus of a specific region, and to take into account common environmental laws put in place by the European Union. This could for instance be the EU taxonomy. The EU taxonomy gives firms and investors clear definitions of which activities that can be thought about as environmentally sustainable. This is one of the ways, which helps firms and investors become more climate friendly and to protect investors, who seek to invest in sustainable firms, from greenwashing. Greenwashing is when firms or organisations claim that they have positive environmental effect without having any evidence to back it up. The EU taxonomy can therefore help the market to shift the investments to where they're needed the most, and help investors avoid greenwashing.

2 European context

Europe has, as one of the wealthiest regions in the world, historically been one of the primary polluters of the world. This pollution, due to industrialization, has been one of the main drivers towards climate change. As climate change is a high priority for many individuals and political parties, it would therefore be interesting to see, if there actually is a correlation between the ESG-score and support towards the "greener" firm in form of a higher debt ratio in Europe.

Since 1973 the European Union has put out legislative proposals and goals for EU environment policy. In May 2022, the EU made yet another agenda for the environment policy, which is in force until the end of 2030. Some of the new proposals and goals are that the EU want to achieve climate neutrality by 2050, the ambition of zero-pollution for air, water and soil and the reduction of environmental and climate pressures related to production and consumption in industries such as energy, industrial development building and infrastructure.[\[1\]](#) [\[2\]](#) In this paper, I will measure how "green" or environmentally friendly a company is by their ESG-score.

2.1 ESG-Scores

ESG stands for Environmental, Social and Governance. All three categories are split into several sub-categories, as seen in Table [1](#), where some of them are stated.

Environmental	Social	Governance
Climate change	Human capital	Corporate governance
Natural resources	Product liability	Corporate behavior
Pollution & waste	Stakeholder opposition	
Environmental opportunities	Social opportunities	

Table 1: MSCI ESG Research’s Key Issue Hierarchy [3](#)

ESG-scores have been around for several decades, but have in recent years gained attention and popularity among investors seeking to have environmental, social and governance factors integrated into their decision-making when investing. It is possible to get ESG-scores from multiple rating agencies such as MSCI and Moody’s. Corporate engines, such as Bloomberg or Refinitiv Eikon, does also provide their own ESG ratings. I’ve found, that the majority of ESG-scores are hidden behind a pay-wall or subscription-based services. Investors wanting to use ESG-scores in their investment decision, will therefore need to pay for it. This can thus limit the usage of ESG-scores, as they might not be as available to the individual investor, but more available to corporations. As rating agencies prioritise different sectors and regions differently, different agencies might provide different ESG-scores. The individual ESG-scores are therefore not perfect, and investors wanting to minimize the imperfections of the ESG-score, could use the average taken from multiple rating agencies or corporate engines.

This is partly due to the way the importance of the ESG factors varies depending on the specific investment strategy or industry. The weights of ESG factors are therefore a matter of judgement and is depended on the specifics of the investment. It is therefore important to note how different factors are weighted and which factors are used, in order to make an informed decision.

For this paper, and the analysis, I will be focusing on the overall ESG-score. But, all aspects of the ESG factors have some important intuition that could be interesting to focus on. The environmental factor is of importance, as firms who are relying on fossil fuels is at risk of new regulations. This can also be seen in Table [1](#) that some of the categories in the environmental pillar are pollution & waste and climate change. The social factor gives insight to firms impact on society. Firms with higher social scores are better managed, more sustainable and give better opportunities for

shareholders. In Table [1](#) we see, that some factors in the social pillar are human capital and social opportunities. The governance factor could also be interesting to look at separately, to see if firms with weak governance practice will be more like to have an excessive amount of debt. This would follow along with the factors in the governance pillar, which includes corporate behavior.

A more practical way of implementing sustainable goals to a firm is via the United Nations' sustainable development goals. These goals are not binding, but they are valuable for firms, as each ESG factor roughly can be translated to one of the 17 sustainable development goals.

3 Literature review

In order to be able to discuss and analyze how ESG-scores could possible effect the debt-to-equity ratio, I first need to know about what determines capital structure. Modigliani and Miller(1958)[\[4\]](#) argues that with perfect capital markets, that the capital structure is irrelevant to determining the overall value of a firm. Instead, they suggest that it is the firms investment decisions that determine the value of a firm. The paper didn't state anything about benefits from debt, which Modigliani and Miller(1963)[\[5\]](#) later corrected by recognizing the advantage of debt due to the tax-shield. This advantages was further looked at by Kraus and Litzenberger(1973)[\[6\]](#), who found that a high leverage causes higher cost of financial distress which could offset the benefits gained from the tax-shield. This became to be known as trade-off theory, where a firm should find their optimal debt level by maximizing the advantages from debt tax-shields and which minimizes the possibility of bankruptcy.

As I will be focusing on the relationship between ESG-scores and the debt-ratio of firms, I will draw some inspiration from the research done by Apergis et. al.(2022)[\[7\]](#). They have done research about the relation between the cost of debt and ESG scores. They show evidence, that the cost of debt for borrowing firms is being affected by their ESG score. They show, that firms with a low ESG score are considered riskier, as they are exposed to liabilities related to the ESG factors. This increases their probability of default. They use a framework to evaluate ESG performance where they investigate firms within S&P500.

As green investments seems to be considered less risky, Raimo et. al.(2021) [\[8\]](#) look at the benefits of ESG disclosure. They find a negative effect between ESG disclosure and the cost of debt financing. This tells us, that the greater the ESG disclosure, the more a firm might benefit from third party financial resources at better conditions. These better conditions could include lower interest rates or longer repayment terms.

Admati et al.(2022)[\[9\]](#) write about the leverage ratchet effect. The effect is a po-

tential factor, that can affect the capital structure choices firm may engage in. The leverage ratchet effect suggest that firms may be less willing to reduce their leverage in times of economic expansion, due to concerns of missing out on profits. And that they may be forced for lower their leverage during times of economic stress, which could impact their ability to finance investments with debt. This effect can also be expected to be seen with firms wanting to finance green investments.

An important observation was made by Fernández-Cuesta et. al .[\[10\]](#). They found, that a better carbon performance allowed industrial firms to obtain a more long-term debt to finance their environmental investments, based on data from the financial leverage of listed firm included in the main stock index of 16 European countries. They get to this result, by using panel data regression, as they analyze the impact of firms reduction of carbon emission on the financial leverage.

4 Debt financing

The overall capital structure of a firm, which refers to the proportion of debt and equity used for financing, has been argued by Modigliani and Miller[\[4\]](#) to be irrelevant to the firms value. The way a firm chooses to structure their capital thus comes with its advantages and disadvantages. An advantages is the tax-shield. This can lead to a reduction in the amount of taxes paid. On the other hand, acquiring more debt increases the probability of default, as the firm will have higher interest or principal payments.[\[6\]](#)

The borrowing cost of a firm is determined by the interest rate paid on debt obligations. This interest rate depends on factors such as a firms credit rating and general market conditions. Credit rating agencies such as Moody's Investor Services or Standard and Poor's evaluate the creditworthiness of borrowers. A higher rating indicates a good borrower, which leads to better interest rates.

Debt maturity, which is represented as the duration of debt obligations, can affect both default risk and borrowing costs. A long debt maturity can reduce the default risk but may also result in higher borrowing costs, while short debt maturity can decrease default risk but also result in an increase in the probability of default. The default probability refers to the likelihood that a firm will default on its debt obligations. This probability is partly determined by factors such as the credit rating and the amount of debt obligations. As written in the literature review, a high leverage causes higher cost of financial distress.[\[6\]](#) This means that there will be higher costs to the firm if they're not able to pay interest payments on their obligations.

5 Hypothesis development

The first hypothesis is going to be about how the ESG score affect the debt-to-equity ratio. On one hand you could argue that firms with a high ESG score are seen as more responsible and sustainable for investors. This could lead to an increased demand for their debt, which will then be increased, as one could expect lower borrowing costs. Firms with higher ESG score might also have more greener financing options, which might lead to them investing in more projects and then acquiring more debt. On the other hand, firms with higher ESG scores can be viewed as less risky and more with more long term prospect, as found by Apergis et. al(2022)[7] and Fernández-Cuesta et. al.[10]. This could lead to firms being able to finance themselves at better conditions. This could then mean that the firms with high ESG scores doesn't need more debt to finance future projects.

As I've now describe why the ESG score can affect the debt financing towards more and less debt, it can be tempting to just say that the ESG score doesn't affect the debt-to-equity ratio overall. But as we have seen in Apergis et al.(2022)[7], that firms with a low ESG score are considered riskier. In Fernández-Cuesta et. al. [10], we see that firms are able to obtain more long-term debt with a better carbon performance. This carbon performance could possibly be translated into a ESG-score. We have also seen in Raimo et. al.[8] that firms with high ESG disclosure are considered less risky. I will therefore expect firms with high ESG scores to be less risky and more likely to acquire more debt, which then gives us the first hypothesis.

5.1 Hypothesis 1: High ESG scores leads to higher debt-to-equity ratios

Different industries have different debt-to-equity ratio benchmarks, as some industries are more capital-heavy than others. I therefore suspect that the ESG score might have a larger effect on certain industries, which normally is seen as large polluters of greenhouse-gasses. I thereby propose a second hypothesis.

5.2 Hypothesis 2: The effect of ESG scores on the debt-to-equity ratio varies by industry

So, as I expect a high debt-to-equity ratio, I then expect "greener" firms to have less risk connected to them. Following trade-off theory, firms will then have to find their optimal debt level, which maximizes their tax-shield and which minimizes their default costs. I thus expect that the optimal debt level will be higher with a high ESG-score.

6 Empirical Framework

6.1 Data & Method

The data used in this paper is from Refinitiv Eikon. I get data for the STOXX Europe 600 index, which is an index consisting of 600 European firms constructed by STOXX Ltd. STOXX Ltd is a Swiss index provider. I've chosen the STOXX Europe 600 index, as it is a big index which represent the biggest firms in Europe, for which my analysis is based about. As I've chosen an European index, all nominal values used in the analysis is going to be in most used currency in Europe, the Euro. I'm able to get data for 599 out of the 600 firms using Refinitiv Eikons excel add-in. The sample is further reduced due to incomplete data. I end up with a final sample consisting of 483 firms. These firms are split into 11 different sectors based on the Global Industry Classification Standard(GICS). The number of firms in each of the 11 sectors is shown in Figure 1.

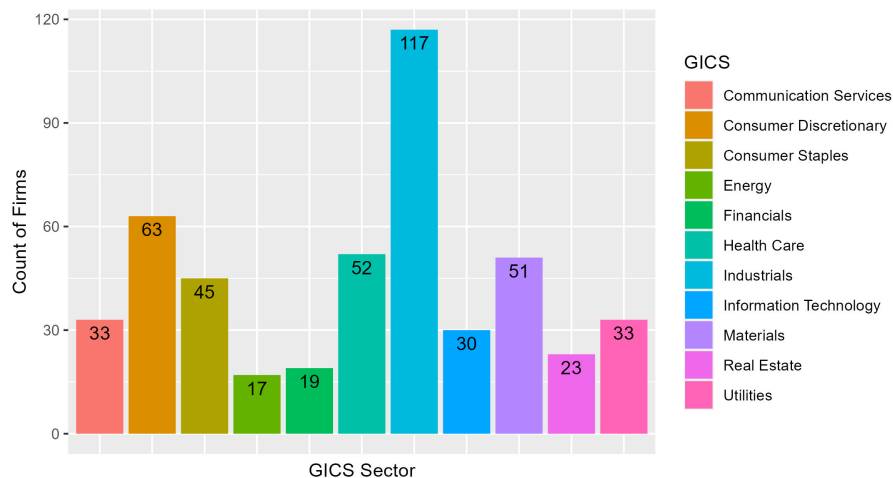


Figure 1: Histogram of number of firms in each sector

In the figure, it is noticeable that there is less than 25 firms in the sectors of Energy(17), Financial(19) and Real Estate(23). We have industrials in the other end of the spectrum with 117 firms, which is almost a fourth of the sample.

6.2 Empirical model

I'm going to construct a model from which I should be able to find the relationship between the ESG-score and the debt-to-equity ratio. I start with the dependent variable being the debt-to-equity ratio and the first parameter being the ESG-score.

$$DE_i = \beta_1 ESG_i + \beta_2 Size_i + \beta_3 ROA_i + \beta_4 AG_i + \epsilon_i \quad (1)$$

I've then used a few control variables in the model in order to better explain the relationship between the debt-to-equity ratio and the ESG-score. Starting with size, I expect it to be positively correlated with the firms debt-to-equity ratio.[\[11\]](#) This is because that larger firms tend to have a more diversified portfolio and thereby a lower default probability. I will be using the log of total assets as the size.

I then add return on assets. The expected sign between the return on assets and the debt-to-equity ratio is uncertain. Following the trade-off theory there's a positive correlation.[\[12\]](#) [\[13\]](#) And following the pecking-order-theory there's a negative one.[\[14\]](#) [\[15\]](#)

Firms with greater growth opportunities have a lower debt ratio.[\[16\]](#) This is due to under-investment problems and that the cost of financial distress are higher for firms with greater growth opportunities. In this paper, I will use asset growth to measure the growth opportunities. Asset growth will be measured as the cumulative capital expenditures, which shows the investments made by a firm in form of .

Variable name	Explanation
DE_i	The Debt-to-equity ratio.
ESG_i	Total ESG-Score from 0 to 100.
$Size_i$	Measured as the log of total assets.
ROA_i	Return on assets.
AG_i	Asset Growth. Calculated as cumulative capital expenditures. Value in millions.
ϵ_i	Error term

Table 2: Explanation of Variables

Regarding the ESG-score, then I've used a score ranging from 0 to 100. As I look at the overall ESG-score, a higher score will thus indicate a better overall performance of the company regarding the environmental, social and governance parameters used to calculate the score. In the case of this paper, the scores are calculated by the companies themselves using a template given by Refinitiv Eikon. This is of course a possible problem, which I will bring up later in the section about limitations.

Looking more specific at what relationship is to be expected regarding the ESG-scores and our variables. Gregory(2022) [17] find that there is a positive relationship between firm size and ESG between agencies. This is also supported by Dremptic et. al(2020) [18] who finds a significant positive correlation between ESG-scores and firm size. Wong et. al(2021) [19] find a positive relationship between ESG and growth opportunities which I interpret as asset growth, even though that they use asset tangibility as a proxy.

6.3 Correlation test

I'll be using the Pearson correlation coefficient measure to test for correlation. The Pearson correlation measures the relationship between two variables. Correlation tells us if the two variables often happen together or not. The values are between -1 and 1 , where any positive value indicates a positive correlation and any negative value indicates a negative correlation. The higher the numerical value, the more the variables correlate. A value of 1 indicates a perfect positive correlation between the two variables. A value of -1 indicates a perfect negative correlation between the two variables. A value of 0 , indicates no correlation between the variables.

The Pearson correlation coefficient between two variables is calculated as the covariance between the two variables over the product of the two individual variables standard deviation.

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \times \sigma_Y} \quad (\text{Pearson correlation coefficient})$$

With $\text{Cov}(X, Y)$ being the covariance between the two variables and σ_X, σ_Y being the standard deviation of X and Y .

I'll be using a significance level of 5% to determine whether or not I can reject the null-hypothesis that there is no correlation between the variables debt-to-equity ratio and ESG-score. I will also be looking at the 1% and 10% significance levels. This is so I'll be able to tell if there is a very significant correlation at a 1% significance level, or if there is only evidence towards a correlation at a 10% significance level.

7 Results

7.1 Descriptive statistics

The descriptive statistics are shown in Table 3, which shows the mean, the standard deviation, the minimum, the maximum and the median for each variable used in the regression. The variables are explained in Table 2.

	Count	Mean	SD	Min	Max	Median
DE	483	88.11	93.37	0.00	991.27	63.07
ESG	483	72.45	13.60	24.01	94.71	74.95
Size	483	22.32	1.59	16.64	27.41	22.30
ROA	483	6.68	11.14	-37.51	186.95	5.82
AG	483	1670.70	3774.94	0.13	33004.00	417.50

Table 3: Summary statistics of the included variables

We get the mean of the debt-to-equity ratio to be 88,11, while it is worth noting, that the minimum debt-to-equity ratio is 0 and the maximum is 991,27. The mean of all the firms ESG-Score is at 72,45. This is close to the median of 74,95, which tells us that there's more firms with a high ESG-score than there is firms with a low ESG-score. Asset growth has the largest mean. And as I use cumulative capital expenditures in millions, we have that the mean of the 483 firms is 1670,70 million. This is a fair bit more than the median, which is at 417,5 million. This thus tell me, that half the firms haven't spent more than 417,5 millions on capital expenditures, and that the firms that have spent more, have spent a lot more. This can be seen as the maximum amount spent is 33.004 million.

7.2 Relationship between ESG and debt-to-equity ratios

Following model (1) I run a regression using an Ordinary Least Squares estimation.¹ The model doesn't pass the RESET test, which is a test to detect potential misspecification of functional form in the model. I then add an additional expression to equation (1). The model now looks like:

$$DE_i = \beta_1 ESG_i + \beta_2 Size_i + \beta_3 ROA_i + \beta_4 AG_i + \beta_5 ESG_i^2 + \epsilon_i \quad (2)$$

Where I've added the squared value of ESG. This model also doesn't pass the RESET test, so the regression model (2) also have a potential misspecification of functional form.² I go back to model (1) and add dummy variables for every GICS sector. From Figure 1, we know that we have 11 different sectors. The model now looks like :

$$DE_i = \beta_1 ESG_i + \beta_2 Size_i + \beta_3 ROA_i + \beta_4 AG_i + \theta_t X_{i,t} + \epsilon_i \quad (3)$$

This model (3) now passes the RESET test, where I have $X_{i,t}$ as the dummy variables for every sector, with their respective coefficient θ_t . All the RESET test specifics are shown in Table 4. I progress with model (3).

	Model (1)	Model (2)	Dummy (3)
RESET	5.0391	4.9325	2.8528
P-Value	0.0068	0.0076	0.0587

Table 4: Misspecification results

Model (3) passes visual misspecification tests, which can be seen in Figure 3. Here it can be seen that the residuals seems to follow the line, which supports the OLS the assumption of normality. It does not pass the test for heteroscedasticity, which can be visually seen in Figure 2. To be sure that it is the case, I also do a White-test. Here I get a P-value extremely close to 0.³ This shows that there might be heteroscedasticity. I therefore proceed with the same model, but with robust standard errors. I use robust standard errors to make sure that the OLS estimators are efficient. The OLS estimator would thus be inefficient with normal standard errors with the presence of heteroscedasticity. The regression results are showed in Table 5.

As seen in Table 5 we have the regression result with 10 dummy variables for

¹The regression results can be found in Table 8

²The regression results can be found in Table 8

³p-value < 0.00000000000000022

	(3)
ESG	0.5649* (0.3422)
Size	-0.6193 (3.3840)
ROA	-1.1521*** (0.4371)
AG	0.0006 (0.0015)
Consumer Discretionary	-9.1365 (18.2894)
Consumer Staples	-26.8200* (13.7798)
Energy	-29.7807 (30.4210)
Financials	-35.6004 (21.9202)
Health Care	-39.4385** (16.6459)
Industrials	-17.1384 (16.4260)
Information Technology	-68.6306*** (13.2495)
Materials	-61.7307*** (13.4197)
Real Estate	-34.0907** (16.0528)
Utilities	65.8465*** (21.9508)
Constant	90.2106 (70.3212)
Observations	483
R ²	0.1432
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

Table 5: Regression results with dummies, robust standard errors

the sectors. In total, we have 11 sectors. The sector "Communication Services" is used as a reference sector for all sector dummy variables.

The parameter of most importance to this paper is the ESG-score. I find a positive coefficient of 0.5649 with a p-value of less than 10%. This is in line with hypothesis 1, which states that a high ESG-score leads to a higher debt-to-equity ratio. It is significant on a 10% level, and not on a 5% level or less. It is thus not supported by the normal 5% significance, which is wanted, but the 10% significance shows us that there is some evidence that point in the right direction.

I also find the ROA to be significantly negative, with a coefficient of -1.1521 with a p-value lower than 0.01. This supports the pecking-order-theory. [14] [15] The other control variables, size and asset growth, both have coefficients which are not significant. The coefficient of Size is -0.6193 with a robust standard deviation of 3.3840. I expected size to be positive, which it isn't. Its negative sign is not significant. The same can be said about asset growth, which I expected to be negatively correlated with the debt-to-equity ratio, but in this case basically looks to be a coefficient of 0.

Table 6: Correlation Matrix, All Sectors

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.0844*	1.0000			
Size	0.0832*	0.3419***	1.0000		
ROA	-0.1673***	-0.0378	-0.1748***	1.0000	
AG	0.0805*	0.2197***	0.5368***	-0.0135	1.0000
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01		

The Pearson correlation coefficient matrix for all sectors are showed in Table 6. In this table, we see the correlation coefficients for all the variables used in the regression, with their significance level. Noticeably, I find that there is a slight positive correlation between the debt-to-equity ratio and the ESG-score. This slight positive correlation is only significant on a 10% level, so it is not supported by the normal 5% significance, which is wanted. Though, as I find the correlation to be significant on a 10% level, there is some evidence which could indicate that there is a positive linear relationship between the debt-to-equity ratio and the ESG-score.

If we keep looking at the ESG-scores, it is also noticeable that there is a positive correlation between ESG-scores and size and asset growth. Both are significant on a 1% significance level. The hypothesis is not focused around this, but I think it is worth noting that ESG-scores and asset growth are slightly positively correlated. Table 6 also shows that there is a negative correlation between the debt-to-equity ratio and the return on asset. In fact, the negative correlation of -0.1676 is significant on a 1% level, which again supports the pecking-order theory. [14] [15]

7.3 Variation by industry

As I've covered most of the analysis, I still need to look into if there are different effects of ESG on the debt-to-equity ratio per sector. In Table 7 I've put the correlation between the debt-to-equity ratio and ESG-scores. The full correlation matrix

per sector can be found in the appendix.

	COM	CD	CS	ENE	FIN	HEA
Corr	0.3736	-0.1031	0.2844	-0.1490	0.4101	0.1045
	IND	INF	MAT	REA	UTI	
Corr	0.1096	0.0476	0.0971	-0.3227*	0.2142	
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01			

Table 7: Correlation between DE and ESG per sector

Table 7 shows that no sectors have a significant correlation with respect to the debt-to-equity ratio and the ESG-score on a 5% significance level. And only the sector "Real Estate" show evidence of negative correlation at a 10% significance level. This doesn't tell me that much, so instead I proceed to look at the regression result in Table 8, which has every sector as a dummy variable. That is, as mentioned earlier, besides "communication services" which is used as a reference sector.

So when we get that "Utilities" has a coefficient of 65.8465 with a p-value of less than 1%, then it simply means that firms within the utilities sector has a debt-to-equity ratio increase of 65.8465 while holding other variables constant. That is, an increase compared to the reference sector. And in fact, the sector "Utilities" is the only sector which gives an increase in the debt-to-equity ratio. All other sectors therefore has a negative coefficient. It is therefore "Utilities" and the reference sector "Communication Services", which has the highest debt-to-equity ratio. In the other end, we have "Information Technology" with the lowest, being at -68.6306 with a p-value lower than 0.01. The difference between the highest of lowest debt-to-equity ratio is 134.4771 and is between "Utilities" and "Information Technology". This shows that there is a general difference between the debt-to-equity level of different sectors, but it doesn't tell anything about if it is because of the ESG score.

8 Discussion

There is a few limitations in this paper. One of them is the basis of ESG-scores. ESG-scores are different depending on who makes them. This is due to difference in methodologies, data sources and areas of focus. Due to this, investors will be better off using multiple sources of ESG data. This will give them a more comprehensive picture of a firms overall sustainability and ethical practices. The ESG-scores used

for this paper are not based on multiple sources. Instead, I've only used the one from Refinitiv Eikon, which are based on the firms own evaluation. I would instead propose to use a different set of ESG-scores from a different platform or an average from multiple sources for future research.

Given that I had to specify the regression model twice, and that I ended up with a model full of dummy variables, it is convenient to talk about the potential addition of more control variables. The final models addition of dummy-variables could possibly have been avoided by adding more control variables to the initial model. Though, I feel that the variables used in the initial model cover a good portion of what could be expected to influence the debt-to-equity ratio, which is why I initially didn't add more control variables.

8.1 Comparison to previous literature

From Fernandez-Cuesta(2019)[\[10\]](#) we know that firms with better carbon performance have more long-term debt. Those terms can roughly be translated to firms with higher ESG-scores have more debt. I find evidence of the same, as my regression shows that the debt-to-equity ratio increases with a higher ESG-score. Though only on a 10% significance level, I still find something similar.

I also find that there's a positive correlation between ESG-scores and asset growth. This is supported by Wong et. al(2021)[\[19\]](#). This indicates that firms with higher ESG-scores have greater growth opportunities. From Myers(1977)[\[16\]](#) we know that firms with greater growth opportunities have a lower debt ratio. This might be one of the reasons why I don't see the ESG-score having a larger or more significant coefficient or correlation, as this effect draws in the other direction. I also find a positive correlation between ESG-scores and firm size. This is also found by Gregory(2022)[\[17\]](#) and Dremptec et. al(2020)[\[18\]](#).

From literature there is uncertainty about the correlation of the debt-to-equity ratio and the return on asset. In this paper, with my data and my specified model, I find a negative coefficient and negative correlation, both with a significance level of 1%. These results are in line with what Myers(1984)[\[14\]](#) and Fama French(2002)[\[15\]](#) suggests.

8.2 Further research

As mentioned previously, there is room for further research regarding the use of different ESG-scores from different or multiple sources. Further research could investigate if there is a difference between different ESG-score sources and possibly use an average of multiple sources.

Due to the limited amount of pages allowed for this paper, I haven't been able to do separate regression for each sector. The 11 sectors would need to have separate regression models, where all 11 would have to be specified. This would take quite a lot of space, and I would therefore propose to do it at another time. It could also be interesting to do the main regression on other indexes in other regions of the world to see if the impact of ESG-scores on the debt-to-equity ratio is the same.

9 Conclusion

The objective of this paper was to investigate whether or not ESG-scores has an effect on the debt-to-equity ratio. Previous literature has focused on ESG disclosure and the relationship between ESG-scores and the cost of debt. My research contribute to existing literature, by looking at a different relationship. My first hypothesis, which is stating that high ESG-scores leads to higher debt-to-equity ratio cannot be confirmed on a 5% significance level, which is academia standard. Though, it can be confirmed on a 10% significance level, which provides evidence to say that ESG-scores potentially could have an effect on the debt-to-equity ratio. It might be beneficial for further analysis to add more control variables in the model or to add more firms in the sample. By expanding the analysis to include more firms, the relationship could become more or less significant, which could give a better answer to whether or not ESG-scores has an effect on the debt-to-equity ratio. I also find that there is a very significant positive correlation between ESG-scores and size and between ESG-scores and asset growth. Both these positive correlations could be a prospect for further research in the future, where it might provide a clearer picture of how ESG-scores affects all aspects of a firm.

My second hypothesis states that the effect of ESG scores on the debt-to-equity ratio varies by industry. The results from the analysis points in the direction of dismissing the hypothesis. If I'm only relying on the Pearson correlation coefficients, then I would dismiss the hypothesis, as no sector is significant on a 5% level and only 1 sector is significant on a 10% level. The main regression does show, that different sectors have different debt-to-equity ratios, but it doesn't provide information about if it is because of the ESG-scores. This is an obvious limitation, which should be researched further. Overall, I find evidence that the ESG-scores possibly could have a significant influence on firms debt-to-equity ratio in the STOXX 600 index. I do not find that there is a significant difference between which sector the firm participates in.

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A Regression table

Table 8: Regression results for All GICS sectors

	(1)	(2)
ESG	0.445 (0.328)	0.974 (2.428)
ESG_squared		−0.004 (0.018)
Size	−0.172 (3.308)	−0.061 (3.350)
ROA	−1.379*** (0.384)	−1.380*** (0.384)
AG	0.002 (0.001)	0.002 (0.001)
Constant	66.247 (70.692)	47.262 (111.544)
Observations	483	483
R ²	0.038	0.038
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Other Table

Sector	Abbreviation	Sector	Abbreviation
Communication Services	COM	Industrials	IND
Consumer Discretionary	CD	Information Technology	INF
Consumer Staples	CS	Materials	MAT
Energy	ENE	Real Estate	REA
Financials	FIN	Utilities	UTI
Health Care	HEA		

Table 9: Abbreviation of sectors used in Table 7

B Visual misspecification tests

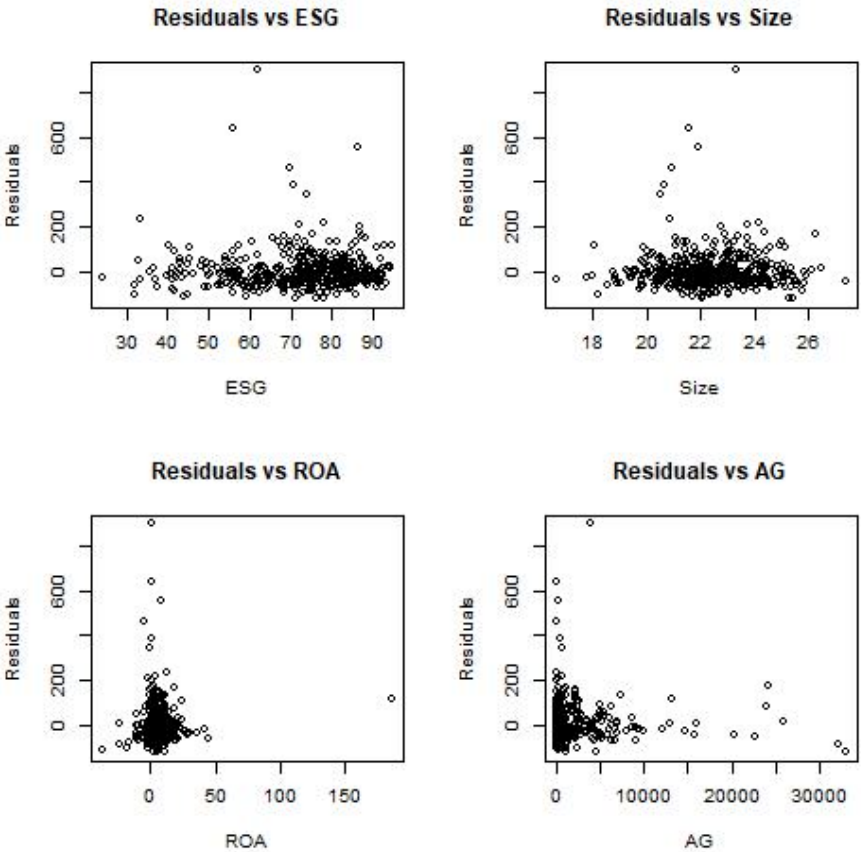


Figure 2: Residual Plots

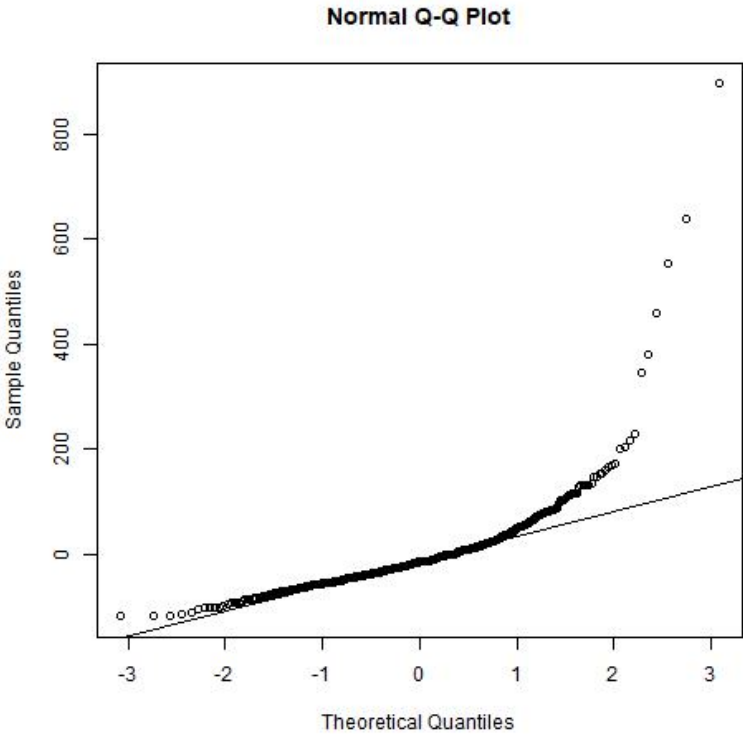


Figure 3: QQ Plot

C Pearson Correlation matrices

Table 10: Correlation Matrix, Communication Services

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.3736	1.0000			
Size	0.5317	0.4253	1.0000		
ROA	-0.2980**	0.0118	-0.4810***	1.0000	
AG	0.6246*	0.3411	0.5941	-0.1343*	1.0000
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01		

Table 11: Correlation Matrix, Consumer Discretionary

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	-0.1031	1.0000			
Size	0.0466	0.5043	1.0000		
ROA	-0.1743	-0.1866	-0.1505	1.0000	
AG	-0.0012	0.2455	0.6368	-0.0452	1.0000
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01		

Table 12: Correlation Matrix, Consumer Staples

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.2844	1.0000			
Size	0.2862	0.3946	1.0000		
ROA	-0.0928	-0.2366	-0.0631	1.0000	
AG	0.2818	0.1772	0.7103*	-0.0393	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 13: Correlation Matrix, Energy

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	-0.1490	1.0000			
Size	-0.3472	0.6306	1.0000		
ROA	-0.4000	-0.3542	-0.0856	1.0000	
AG	-0.2151	0.3321	0.8351**	0.0490	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 14: Correlation Matrix, Financials

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.4101	1.0000			
Size	0.2331	0.2260	1.0000		
ROA	0.0351	0.0597	-0.2959*	1.0000	
AG	-0.0206	-0.1261	0.4224	-0.3390	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 15: Correlation Matrix, Health Care

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.1045	1.0000			
Size	-0.0043	0.5826	1.0000		
ROA	-0.1154	0.1948	0.1226	1.0000	
AG	-0.0465	0.3344	0.6363	0.3165	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 16: Correlation Matrix, Industrials

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.1096	1.0000			
Size	0.0772	0.2852	1.0000		
ROA	-0.1473	-0.1439	-0.1785	1.0000	
AG	0.0999	0.1991	0.5315	0.0156	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 17: Correlation Matrix, Information Technology

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.0476	1.0000			
Size	-0.0063	0.4755	1.0000		
ROA	-0.4123	0.0903	-0.2607	1.0000	
AG	-0.0311	0.4176	0.7561**	0.0514	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 18: Correlation Matrix, Materials

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.0971	1.0000			
Size	-0.3191	0.2298	1.0000		
ROA	-0.1278	-0.3391	-0.1585	1.0000	
AG	-0.3115*	0.2127	0.7581**	0.0298	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 19: Correlation Matrix, Real Estate

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	-0.3227*	1.0000			
Size	0.8236***	-0.2624	1.0000		
ROA	-0.1985	0.1489	-0.4469*	1.0000	
AG	0.2004	0.4172	0.1759	0.0134	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

Table 20: Correlation Matrix, Utilities

	DE	ESG	Size	ROA	AG
DE	1.0000				
ESG	0.2142	1.0000			
Size	-0.1505	0.3962	1.0000		
ROA	-0.5021*	-0.0372	-0.0596	1.0000	
AG	-0.1366	0.1710	0.5795	0.0644	1.0000
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01					

D R-code

The following R-code replicates all tables, figures, misspecificationtests, regressions and correlationtest.

```
#Library
library(dplyr)
library(readxl)
library(tidyverse)
library(tidyquant)
library(RSQLite)
library(slider)
library(scales)
library(furrr)
library(lubridate)
library(sandwich)
library(lmtest)
library(texreg)
library(kableExtra)
library(glmnet)
```

```
library(base)
library(forecast)
library(cowplot)
library(float)
library(ggplot2)
library(tidymodels)
library(timetk)
library(recipes)
library(keras)
library(hardhat)
library(tensorflow)
library(parsnip)
library(ggpubr)
library(xtable)
library(stargazer)
library(knitr)
library(psych)
library(readxl)

#####
#Import of data
#Import excel file
my_data <- read_excel("seminar data_endelig.xlsx")
View(my_data)

my_data <- my_data[complete.cases(my_data),]

#####
#Changing data, names and log
# Renaming columns in my_data
my_data <- my_data %>%
  rename(GICS = 'GICS Sector Name',
         DE = 'Total Debt to Total Equity, Percent',
         ESG = 'ESG Score',
         Size = 'Total Current Assets',
         ROA = 'ROA Total Assets, Percent',
         AG = 'Capital Expenditures, Cumulative'
  )

# Transforming Size to log(Size)
my_data$Size <- log(my_data$Size)
my_data$AG <- my_data$AG * -1
```

```

my_data$AG <- my_data$AG / 1000000

#####
#Plot of the count in different sectors
#(amount of firms in each sector)
ggplot(my_data, aes(x = 'GICS', fill = 'GICS')) +
  geom_bar() +
  labs(x = "GICS Sector",
       y = "Count of Firms") +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text.x = element_blank(),
        axis.ticks.x = element_blank(),
        legend.position = "right")

ggsave
("Histogram of number of firms in each sector.jpg")

#Plot of the count in different sectors
#(amount of firms in each sector)
#With the amount of firms per sector showing on the graph
ggplot(my_data, aes(x = GICS, fill = GICS)) +
  geom_bar() +
  geom_text(stat = "count", aes(label = ..count..),
           vjust = 1.5) +
  labs(x = "GICS Sector", y = "Count of Firms") +
  theme(plot.title = element_text(hjust = 0.5),
        axis.text.x = element_blank(),
        axis.ticks.x = element_blank(),
        legend.position = "right")

ggsave
("Histogram of number of firms in each sector, count.jpg")

#####
#Descriptive statistics
# Load the my_data dataset
data(my_data)
# Set the scipen option to turn off scientific notation
options(scipen = 999)

# Calculate the descriptive statistics for all
# variables in the my_data dataset

```

```

desc_stats <- data.frame(
  count = apply(my_data[c("DE", "ESG", "Size", "ROA", "AG")],
    2, function(x) sum(!is.na(x))),
  mean = apply(my_data[c("DE", "ESG", "Size", "ROA", "AG")],
    2, mean),
  sd = apply(my_data[c("DE", "ESG", "Size", "ROA", "AG")],
    2, sd),
  min = apply(my_data[c("DE", "ESG", "Size", "ROA", "AG")],
    2, min),
  max = apply(my_data[c("DE", "ESG", "Size", "ROA", "AG")],
    2, max),
  median = apply(my_data[c("DE", "ESG", "Size", "ROA", "AG")],
    2, median))

# Add row names to the table
row.names(desc_stats) <- c("DE", "ESG", "Size", "ROA", "AG")

# Print the table with the descriptive statistics
desc_stats

# Set LaTeX options
options(knitr.table.format = "latex")
options(knitr.kable.NA = "")

# Generate the LaTeX table
kable(desc_stats, caption =
"Summary statistics of the included variables",
digits = 2) %>%
  kable_styling(latex_options = "hold_position") %>%
  kable_classic(full_width = F, html_font = "Cambria")

#####
#Regression for the entire dataset
lm_full <- lm(DE ~ ESG + Size + ROA + AG, data = my_data)
#summary(lm_full) # displays the results of the regression
stargazer(lm_full, title =
"Regression results for All GICS sectors", align = TRUE,
type = "latex")

##Misspecification test
library(lmtest)

```

```

resettest(lm_full)
#Doesn't pass

#Adding ESG squared
my_data$ESG_squared <- my_data$ESG^2

#Regression for the entire dataset, with ESG^2
lm_full1 <- lm(DE ~ ESG + ESG_squared + Size + ROA + AG,
data = my_data)
summary(lm_full1) # displays the results of the regression
stargazer(lm_full1, title =
"Regression results for All GICS sectors with ESG-squared",
align = TRUE, type = "latex")

##Misspecification test
resettest(lm_full1)
#Still doesn't pass, trying with dummies for sectors

#####
#With dummy for all sectors
dummy_model <- lm(DE ~ ESG + Size + ROA + AG,
data = my_data)
# Convert GICS to a factor variable
my_data$GICS <- factor(my_data$GICS)
dummy_model <- lm(DE ~ ESG + Size + ROA + AG + GICS,
data = my_data)
#summary(dummy_model)

##Misspecification test
resettest(dummy_model)

#Print table to latex
stargazer(dummy_model, title =
"Regression results with dummies", align = TRUE,
type = "latex")

#More misspecification tests
# Saving the Q-Q plot to a JPEG file
jpeg("qq-plot.jpg")
#Examining the normality of residuals using a Q-Q plot:

```

```

qqnorm(resid(dummy_model))
qqline(resid(dummy_model))
# Seems to follow the line,
# supporting the assumption of normality
# Closing the JPEG file
dev.off()

#Test for homoscedasticity
# Perform the White test
white_test <- bptest(dummy_model, studentize = FALSE)
white_test

#Check for homoscedasticity by plotting residuals
# against the independent variables
# Saving the multi-panel plot as a JPEG file
jpeg("residual_plots.jpg")
# Setting up a multi-panel plot layout
par(mfrow = c(2, 2))
# Plotting residuals against each independent variable
plot(my_data$ESG, resid(dummy_model), xlab = "ESG",
      ylab = "Residuals", main = "Residuals vs ESG")
plot(my_data$Size, resid(dummy_model), xlab = "Size",
      ylab = "Residuals", main = "Residuals vs Size")
plot(my_data$ROA, resid(dummy_model), xlab = "ROA",
      ylab = "Residuals", main = "Residuals vs ROA")
plot(my_data$AG, resid(dummy_model), xlab = "AG",
      ylab = "Residuals", main = "Residuals vs AG")
# Closing the JPEG file
dev.off()

#####
##I now need robust std. errors
# Fit the regression model
dummy_model <- lm(DE ~ ESG + Size + ROA + AG + GICS,
  data = my_data)

# Obtain robust standard errors
robust_se <- sqrt(diag(vcovHC(dummy_model, type = "HC1")))

# Print the coefficients along with robust standard errors
coeftest(dummy_model, vcov = vcovHC(dummy_model,
  type = "HC1"))

```

```

# Print the summary of the model with robust standard errors
summary(dummy_model, robust = TRUE)

# Store the coefficients and robust standard errors
coefficients_table <- coeftest(dummy_model,
vcov = vcovHC(dummy_model, type = "HC1"))
coefficients <- coefficients_table[, 1]
robust_standard_errors <- coefficients_table[, 2]

# Create a table using stargazer
stargazer(dummy_model, type = "latex",
  coef = list(coefficients, robust_standard_errors),
  se = list(robust_se),
  title = "Regression results with dummies,
robust standard errors",
  header = FALSE,
  digits = 4,
  align = TRUE)

#####
# Calculate the Pearson correlation coefficient for the
#variables used in the regression
cor_matrix <- cor(my_data[c("DE", "ESG", "Size", "ROA",
"AG")], method = "pearson")

show(cor_matrix)

# Get the p-values for the correlation coefficients
# Specify the variables I want to calculate correlations
variables <- c("DE", "ESG", "Size", "ROA", "AG")

# Create an empty table to store the correlation
# coefficients and p-values
corr_table <- matrix(0, nrow = length(variables),
ncol = length(variables))
colnames(corr_table) <- variables
rownames(corr_table) <- variables

# Create an empty table to store the p-values
pvalue_table <- matrix(0, nrow = length(variables),
ncol = length(variables))

```

```

colnames(pvalue_table) <- paste0(variables, "_pvalue")
rownames(pvalue_table) <- variables

# Calculate correlation coefficients and p-values
for (i in 1:(length(variables)-1)) {
  for (j in (i+1):length(variables)) {
    result <- cor.test(my_data[[variables[i]]],
                      my_data[[variables[j]]], method = "pearson")
    corr_table[i, j] <- result$estimate
    corr_table[j, i] <- result$estimate
    pvalue_table[i, j] <- result$p.value
    pvalue_table[j, i] <- result$p.value
  }
}

# Combine the correlation and p-value tables
corr_pvalue_table <- cbind(corr_table, pvalue_table)

# Print the correlation table with significance levels.
kable(corr_pvalue_table, format = "latex")
# Bad table, better with only the p-values
# Print the p-value table
kable(pvalue_table, format = "latex")

#####
# Now I need the correlation matrix for each GICS sector
# I start by calculating the correlation coefficient
# Define the variables used in the regression
vars <- c("DE", "ESG", "Size", "ROA", "AG")

# split my_data by GICS sector
my_data_by_sector <- split(my_data[vars], my_data$GICS)

# calculate the Pearson correlation coefficient
# for each sector separately
cor_by_sector <- lapply(my_data_by_sector, function(x) {
  cor_matrix <- cor(x, method = "pearson")
  return(cor_matrix)
})

cor_by_sector[["Communication Services"]]

```

```

cor_by_sector[["Consumer Discretionary"]]
cor_by_sector[["Consumer Staples"]]
cor_by_sector[["Energy"]]
cor_by_sector[["Financials"]]
cor_by_sector[["Health Care"]]
cor_by_sector[["Industrials"]]
cor_by_sector[["Information Technology"]]
cor_by_sector[["Materials"]]
cor_by_sector[["Real Estate"]]
cor_by_sector[["Utilities"]]

#####
# Now I get the p-values that goes together with
# the correlation coefficients
# Specify the variables you want to calculate
# correlations for
variables <- c("DE", "ESG", "Size", "ROA", "AG")

# Select the sector you want to analyze
# (e.g., "Communication Services")
sector <- "Communication Services"

# Get the correlation matrix for the selected sector
cor_matrix <- cor_by_sector[[sector]]

# Create an empty table to store the p-values
pvalue_table <- matrix(0, nrow = length(variables),
ncol = length(variables))
colnames(pvalue_table) <- variables
rownames(pvalue_table) <- variables

# Calculate p-values for the selected sector
for (i in 1:(length(variables)-1)) {
  for (j in (i+1):length(variables)) {
    result <- cor.test(cor_matrix[, i], cor_matrix[, j],
method = "pearson")
    pvalue_table[i, j] <- result$p.value
    pvalue_table[j, i] <- result$p.value
  }
}

# Print the p-value table for the selected sector

```

```

#kable(pvalue_table , format = "latex")
print(pvalue_table)

#####
# Saving all the correlation matrices
# Convert the correlation matrix to a data frame
cor_df <- as.data.frame(cor_matrix)
cor_df <- format(cor_df, nsmall = 4)

# Print the correlation matrix to a LaTeX table
print(xtable(cor_df, caption = "Correlation Matrix"),
      caption.placement = "top")

#Saving the correlation matrices
# Create a LaTeX table representation of
# the correlation matrix
cor_table <- xtable(cor_df, caption = "Correlation Matrix,
All")

# Save the table to a file
file_path <- "correlation_table.tex"
# Specify the file path and name
print(cor_table , caption.placement = "top",
      file = file_path)

##### For each sector , separately

#Sector 1
cor_df1 <- as.data.frame(cor_by_sector
[["Communication Services"]])
cor_df1 <- format(cor_df1 , nsmall = 4)
cor_table1 <- xtable(cor_df1 ,
caption = "Correlation Matrix, Communication Services")
# Save the table to a file
file_path <- "correlation_table1.tex"
# Specify the file path and name
print(cor_table1 , caption.placement = "top",
      file = file_path)

#Sector 2
cor_df2 <- as.data.frame(cor_by_sector

```

```
[[ "Consumer Discretionary" ]])
cor_df2 <- format(cor_df2, nsmall = 4)
cor_table2 <- xtable(cor_df2,
caption = "Correlation Matrix, Consumer Discretionary")
# Save the table to a file
file_path <- "correlation_table2.tex"
# Specify the file path and name
print(cor_table2, caption.placement = "top",
file = file_path)

#Sector 3
cor_df3 <- as.data.frame(cor_by_sector
[[ "Consumer Staples" ]])
cor_df3 <- format(cor_df3, nsmall = 4)
cor_table3 <- xtable(cor_df3,
caption = "Correlation Matrix, Consumer Staples")
# Save the table to a file
file_path <- "correlation_table3.tex"
# Specify the file path and name
print(cor_table3, caption.placement = "top",
file = file_path)

#Sector 4
cor_by_sector [[ "Energy" ]]
cor_df4 <- as.data.frame(cor_by_sector [[ "Energy" ]])
cor_df4 <- format(cor_df4, nsmall = 4)
cor_table4 <- xtable(cor_df4,
caption = "Correlation Matrix, Energy")
# Save the table to a file
file_path <- "correlation_table4.tex"
# Specify the file path and name
print(cor_table4, caption.placement = "top",
file = file_path)

#Sector 5
cor_by_sector [[ "Financials" ]]
cor_df5 <- as.data.frame(cor_by_sector [[ "Financials" ]])
cor_df5 <- format(cor_df5, nsmall = 4)
cor_table5 <- xtable(cor_df5,
caption = "Correlation Matrix, Financials")
# Save the table to a file
file_path <- "correlation_table5.tex"
```

```
# Specify the file path and name
print(cor_table5, caption.placement = "top",
file = file_path)

#Sector 6
cor_by_sector[["Health Care"]]
cor_df6 <- as.data.frame(cor_by_sector[["Health Care"]])
cor_df6 <- format(cor_df6, nsmall = 4)
cor_table6 <- xtable(cor_df6,
caption = "Correlation Matrix, Health Care")
# Save the table to a file
file_path <- "correlation_table6.tex"
# Specify the file path and name
print(cor_table6, caption.placement = "top",
file = file_path)

#Sector 7
cor_by_sector[["Industrials"]]
cor_df7 <- as.data.frame(cor_by_sector[["Industrials"]])
cor_df7 <- format(cor_df7, nsmall = 4)
cor_table7 <- xtable(cor_df7,
caption = "Correlation Matrix, Industrials")
# Save the table to a file
file_path <- "correlation_table7.tex"
# Specify the file path and name
print(cor_table7, caption.placement = "top",
file = file_path)

#Sector 8
cor_by_sector[["Information Technology"]]
cor_df8 <- as.data.frame(cor_by_sector
[["Information Technology"]])
cor_df8 <- format(cor_df8, nsmall = 4)
cor_table8 <- xtable(cor_df8,
caption = "Correlation Matrix, Information Technology")
# Save the table to a file
file_path <- "correlation_table8.tex"
# Specify the file path and name
print(cor_table8, caption.placement = "top",
file = file_path)

#Sector 9
```

```
cor_by_sector[[" Materials "]]
cor_df9 <- as.data.frame(cor_by_sector[[" Materials "]])
cor_df9 <- format(cor_df9, nsmall = 4)
cor_table9 <- xtable(cor_df9,
caption = "Correlation Matrix, Materials")
# Save the table to a file
file_path <- "correlation_table9.tex"
# Specify the file path and name
print(cor_table9, caption.placement = "top",
file = file_path)

#Sector 10
cor_by_sector[[" Real Estate "]]
cor_df10 <- as.data.frame(cor_by_sector[[" Real Estate "]])
cor_df10 <- format(cor_df10, nsmall = 4)
cor_table10 <- xtable(cor_df10,
caption = "Correlation Matrix, Real Estate")
# Save the table to a file
file_path <- "correlation_table10.tex"
# Specify the file path and name
print(cor_table10, caption.placement = "top",
file = file_path)

#Sector 11
cor_by_sector[[" Utilities "]]
cor_df11 <- as.data.frame(cor_by_sector[[" Utilities "]])
cor_df11 <- format(cor_df11, nsmall = 4)
cor_table11 <- xtable(cor_df11,
caption = "Correlation Matrix, Utilities")
# Save the table to a file
file_path <- "correlation_table11.tex"
# Specify the file path and name
print(cor_table11, caption.placement = "top",
file = file_path)
```